

OS Unit 2 Previous Year Questions

Operating Systems (OS) · Unit 2 · CPU Scheduling & Synchronization · 2025

Questions collected from previous year university examinations (2021–2025).

Part A — Short answer (2 marks)

1. Compare Scheduling Criteria with FCFS, SJF and Round Robin. When would you prefer one over the other? [2023]
2. Compare FCFS, SJF and Round Robin with Critical Section Problem. When would you prefer one over the other? [2022]
3. Explain Priority Scheduling with a neat diagram and a suitable example. [2023]
4. Compare Critical Section Problem with FCFS, SJF and Round Robin. When would you prefer one over the other? [2022]

Part B — Long answer (10 marks)

1. Define Semaphores. Derive or justify the key properties involved. [2025]
2. Explain Classic Synchronization Problems with a neat diagram and a suitable example. [2025]
3. Compare Scheduling Criteria with Critical Section Problem. When would you prefer one over the other? [2022]
4. Discuss the role of FCFS, SJF and Round Robin in CPU Scheduling & Synchronization. [2024]
5. Write short notes on Priority Scheduling. [2022]
6. What is meant by Critical Section Problem? Discuss its advantages and limitations. [2021]
7. Compare Semaphores with Scheduling Criteria. When would you prefer one over the other? [2023]
8. Discuss the role of Classic Synchronization Problems in CPU Scheduling & Synchronization. [2021]
9. Define Scheduling Criteria. Derive or justify the key properties involved. [2022]
10. Write short notes on FCFS, SJF and Round Robin. [2025]